Niladri Pradhan

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Professional Summary

Results-driven **AI Engineer** and BCA graduate with hands-on expertise in **Computer Vision, Machine Learning, and Deep Learning**. Proficient in designing and deploying end-to-end AI solutions using **YOLOv8, TensorFlow, OpenCV, and Python**. Demonstrated ability to build real-time object detection, OCR, and NLP-based systems with measurable business impact. Seeking a challenging role as an **AI Engineer, Machine Learning Engineer, Computer Vision Engineer,** or **Data Scientist** to contribute to innovative AI-driven products.

Technical Skills

Languages & Frameworks: Python, SQL

AI / ML & Deep Learning: Machine Learning, Deep Learning, CNNs, TensorFlow, YOLOv8, scikit-learn

Computer Vision & OCR: OpenCV, EasyOCR, Image Processing, Real-Time Object Detection

Data Science: Data Analysis, Feature Engineering, Model Evaluation, Data Preprocessing

Tools & Platforms: Git, GitHub, Jupyter Notebook, VS Code, MySQL

Projects

License Plate Recognition System | *YOLOv8, EasyOCR, Python, SQL, OpenCV* **2025**

- Engineered a **real-time vehicle license plate detection pipeline** using YOLOv8 and EasyOCR, processing both live video streams and static images with high precision.
- Achieved **90%+ character-level accuracy** in license plate text extraction under varied lighting and angle conditions through model fine-tuning and image pre-processing.
- Designed and integrated an **automated SQL database module** to persist extracted vehicle data, enabling structured querying and audit trail creation.
- Reduced manual vehicle logging effort by **simulating a fully automated entry/exit logging system**, applicable to parking management and traffic surveillance.

PPE Safety Detection System | *YOLOv8, OpenCV, Python, Deep Learning* **2025**

- Architected an **AI-powered safety compliance monitoring system** leveraging a custom-trained YOLOv8 model to detect Personal Protective Equipment (PPE) in real time.
- Trained the model to **classify helmets and safety vests** across diverse industrial environments with bounding-box precision, processing video feeds at **25+ FPS**.
- Implemented an **automated real-time alert system** that triggers instant notifications upon PPE violation detection, improving workplace safety enforcement.
- Deployed the pipeline in a simulated industrial environment, demonstrating potential to **reduce safety incidents by enabling zero-delay violation reporting**.

Resume Screening AI | *Python, NLP, Machine Learning, scikit-learn, SQL* **2025**

- Developed an **end-to-end ML-based resume ranking engine** that parses resumes, extracts key skills via NLP, and ranks candidates against job descriptions using cosine similarity.
- Engineered a **skills-matching algorithm** that maps extracted candidate competencies to job requirements, producing a relevance score to assist recruiters in shortlisting.
- Automated the candidate shortlisting workflow, **reducing manual resume review time by an estimated 70%** for a pipeline of 100+ resumes.
- Integrated structured data output to facilitate downstream filtering and storage, enabling **scalable, bias-reduced candidate evaluation**.

Education

Bachelor of Technology (B.Tech) in Information Technology **2022 – 2025**
Seacom Engineering College *Howrah, West Bengal*

Diploma in Computer Science and Technology **2020 – 2022**
Central Calcutta Polytechnic *Kolkata, West Bengal*